1. (12 pts)

4 PTS (a) Find the area of the triangle through the points $A(1,2,1)$, $B(3,1,2)$, and $C(2,4,3)$.

$+1 \left\{ \begin{array}{l} \vec{AB} = \langle 2,-1,1 \rangle \\ \vec{AC} = \langle 1,2,2 \rangle \end{array} \right.$

$+2 \left\{ \begin{array}{l} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 1 \\ 1 & 2 & 2 \end{vmatrix} \\ = (-2-2)\vec{i} - (4-1)\vec{j} + (4-(-1))\vec{k} \\ = \langle -4,-3,5 \rangle \end{array} \right.$

$\left. \begin{array}{l} \text{AREA} = \frac{1}{2}\sqrt{(-4)^2+(-3)^2+(5)^2} \\ = \frac{1}{2}\sqrt{16+9+25} \end{array} \right\} +1$

Area = $\frac{1}{2}\sqrt{50} = \frac{5}{2}\sqrt{2}$

4 PTS (b) Consider the curve $y = -x^2 + 4x + \sin(x-1)$. At the point where $x = 1$, give a unit vector that is parallel to the tangent line and a unit vector perpendicular to the tangent line.

$+1 \left\{ \begin{array}{l} y' = -2x+4+\cos(x-1) \\ y'|_{x=1} = -2+4+\cos(0) = 3 \end{array} \right.$

A VECTOR PARALLEL TO TANGENT LINE: $\langle 1,3 \rangle$ OR $\langle -1,-3 \rangle$ } +1

A VECTOR PERPENDICULAR TO THE TANGENT LINE: $\langle 3,-1 \rangle$ OR $\langle -3,1 \rangle$ } +1

+1 { LENGTH = $\sqrt{1+9} = \sqrt{10}$

ACCEPT EITHER

a unit vector parallel to tangent = $\frac{1}{\sqrt{10}}\langle 1,3 \rangle$ OR $\frac{-1}{\sqrt{10}}\langle 1,3 \rangle$

a unit vector perpendicular to tangent = $\frac{1}{\sqrt{10}}\langle 3,-1 \rangle$ OR $\frac{1}{\sqrt{10}}\langle -3,1 \rangle$

4 PTS (c) Find the center and radius of the sphere with P such that the distance from P to $(0,0,4)$ is triple the distance from P to $(0,0,0)$.

$+1 \left\{ \sqrt{x^2+y^2+(z-4)^2} = 3\sqrt{x^2+y^2+z^2} \right.$

$+1 \left\{ \begin{array}{l} x^2+y^2+z^2-8z+16 = 9(x^2+y^2+z^2) \\ x^2+y^2+z^2-8z+16 = 9x^2+9y^2+9z^2 \end{array} \right.$

$+1 \left\{ \begin{array}{l} 16 = 8x^2+8y^2+8z^2+8z \\ 2 = x^2+y^2+z^2+z \end{array} \right.$

$\frac{1}{2} \rightarrow (\frac{1}{2})^2$

$+1 \left\{ \Rightarrow 2+\frac{1}{4} = x^2+y^2+(z+\frac{1}{2})^2 \right.$

Center = $(0,0,-\frac{1}{2})$

Radius = $\sqrt{9/4} = 3/2$

2. (14 pts)

$\boxed{2\text{ PTS}}$ (a) Assume **a**, **b** and **c** are vector in 3-dimensions. For each of the following expressions write if the result would be 'a Vector', 'a Scalar', or 'Meaningless'.

+1 i. $(\mathbf{a}+\mathbf{b})\cdot\mathbf{c}$ is a SCALAR

+1 ii. $\mathbf{a}\times\mathbf{b}-\mathbf{c}$ is a VECTOR

$\boxed{6\text{ PTS}}$ (b) Find parametric equations for the line of intersection of the planes $x+2y-z=4$ and $2x-y+z=1$.

$$+2\left\{\begin{array}{l} \begin{cases} x+2y-z=4 \\ 2x-y+z=1 \end{cases} \quad z=1-2x+y \\ 3x+y=5 \\ P(\dot{0},5,6) \quad x=0 \Rightarrow y=5 \Rightarrow z=1-0+5=6 \end{array}\right.$$

$$+2\left\{ Q(\dot{1},2,1) \quad x=1 \Rightarrow 3+y=5 \Rightarrow y=2 \Rightarrow z=1-2+2=1 \right.$$

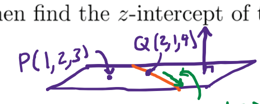
$$+2\left\{ \overrightarrow{PQ}=\langle 1,-3,-5\rangle \right.$$

WE ACCEPT ANY ANSWER WHERE
(1) DIRECTION VECTOR IS PARALLEL TO $\langle 1,-3,-5\rangle$
(2) POINT USE IS ON BOTH PLANES

Line Equation: $x=t,\ y=5-3t,\ z=6-5t$

$\boxed{6\text{ PTS}}$ (c) Find the equation of the plane that passes through the point $(1,2,3)$ and contains the line $x=2+t$, $y=1+t$, $z=4+2t$. Then find the z-intercept of this plane.

GIVEN: $P(1,2,3)$, $Q(2,1,4)$ ARE ON THE PLANE



+1 $\longrightarrow$ $\langle 1,1,2\rangle$ IS PARALLEL TO THE PLANE $\langle 1,1,2\rangle$

+1 THUS, $\overrightarrow{PQ}=\langle 1,-1,1\rangle$ AND $\langle 1,1,2\rangle$ ARE BOTH PARALLEL TO THE DESIRED PLANE.

$$+1\left\{ \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 1 \\ 1 & 1 & 2 \end{vmatrix} = (-2-1)\vec{i}-(2-1)\vec{j}+(1-(-1))\vec{k} = \langle -3,-1,2\rangle \right.$$

+2 { Z-INTERCEPT $\Rightarrow x=0, y=0 \Rightarrow -3(0-1)-(0-2)+2(z-3)=0$

$5+2z-6=0$

$2z=1$

$-3(x-1)-(y-2)+2(z-3)=0$

$-3x+3-y+2+2z-6=0$

$-3x-y+2z=1$

WE ACCEPT ANY EQUIVALENT FORM

+1 $\longrightarrow$ Plane Equation: $3x+y-2z=-1$

z-intercept: $(x,y,z) = (0,0,\tfrac{1}{2})$

3. (12 pts)

$-4 = -x^2 + 2y^2 + z^2$

2PTS (a) Give the precise name of the surface given by the equation $4 = x^2 - 2y^2 - z^2$.

ELL, HYP, HYP

Surface name: HYPERBOLOID OF TWO SHEETS

4PTS (b) Find parametric equations for the curve of intersection of the cylinder $x^2 + y^2 = 4$ and the plane $z = x - y$. Then set up, but do **not** evaluate, an integral for the arc length of this curve for one complete trip around the cylinder.

$x = 2\cos(t), y = 2\sin(t), z = 2\cos(t) - 2\sin(t) \quad 0 \le t \le 2\pi$

$x' = -2\sin(t), y' = 2\cos(t), z' = -2\sin(t) - 2\cos(t)$

$$\int_0^{2\pi} \sqrt{\underbrace{4\sin^2(t) + 4\cos^2(t)}_{4} + (-2\sin(t) - 2\cos(t))^2}\, dt$$

ALSO CORRECT $\quad 0 \le t \le 4$

$x = t,\ y = \pm\sqrt{4 - t^2},\ z = t \mp \sqrt{4 - t^2}$

AND THERE ARE OTHERS.

+2 Curve Equations: $\vec{r}(t) = \langle 2\cos(t), 2\sin(t), 2\cos(t) - 2\sin(t) \rangle$

+2 Arc Length Set-Up: $\int_0^{2\pi} \sqrt{4 + 4(\sin(t) + \cos(t))^2}\, dt$

6PTS (c) The curves $\mathbf{r}_1(t) = \langle 2t + 1,\ 6 - t,\ t^2 + 2t + 1 \rangle$ and $\mathbf{r}_2(u) = \langle u - 2,\ u,\ \frac{u+3}{2} \rangle$ intersect at a point. Find the point of intersection and the angle between the curves at that point. Give your final answer for the angle either in simplified exact form or as a decimal rounded to the nearest degree.

+1 INTERSECTION:

$2t + 1 = u - 2$

$6 - t = u$

$t^2 + 2t + 1 = \frac{u+3}{2}$

$\Rightarrow 2t + 1 = (6 - t) - 2$

$2t + 1 = 4 - t$

$3t = 3$

$t = 1$

$\Rightarrow u = 5$

CHECK

$t = 1 \Rightarrow (x, y, z) = (3, 5, 4)$

$u = 5 \Rightarrow (x, y, z) = (3, 5, 4)$ +1

+2 TANGENT DIRECTION:

$\vec{r}_1'(t) = \langle 2, -1, 2t + 2 \rangle \qquad \vec{r}_1'(1) = \langle 2, -1, 4 \rangle$

$\vec{r}_2'(u) = \langle 1, 1, \frac{1}{2} \rangle \qquad \vec{r}_2'(5) = \langle 1, 1, \frac{1}{2} \rangle$

+2 ANGLE $= \cos^{-1}\left(\dfrac{\langle 2, -1, 4 \rangle \cdot \langle 1, 1, 1/2 \rangle}{\sqrt{4 + 1 + 16}\, \sqrt{1 + 1 + 1/4}} \right) = \cos^{-1}\left(\dfrac{2 - 1 + 2}{\sqrt{21}\, \sqrt{9/4}} \right)$

$= \cos^{-1}\left(\dfrac{3}{\sqrt{21}\,(3/2)} \right) = \cos^{-1}\left(\dfrac{2}{\sqrt{21}} \right) \approx 64.123^{\circ}$

ALSO ACCEPT EXACT FORMS

Intersection Point: $(x, y, z) =$ (3, 5, 4)

Angle = 64 degrees

4. (12 pts)

(a) Consider the curve $\mathbf{r}_1(t) = \langle t^2+1,\ 3t-2,\ t^3-t\rangle$.

$\boxed{4\text{pts}}$ i. Find parametric equations for the tangent line to the curve at the point $(2,1,0)$.

$+1$ $\left\{ \begin{array}{l} t^2+1=2 \\ 3t-2=1 \\ t^3-t=0 \end{array} \right\} \Rightarrow t=1$

$+2$ $\left\{ \begin{array}{l} x = 2+2u \\ y = 1+3u \\ z = 0+2u \end{array} \right.$ ACCEPTABLE TO USE t HERE

$+1$ $\left\{ \begin{array}{l} \vec{r}\,'(t) = \langle 2t, 3, 3t^2-1\rangle \\ \vec{r}\,'(1) = \langle 2,3,2\rangle \end{array} \right.$

Tangent Line Equations: $x=2+2u,\ y=1+3u,\ z=2u$

$\boxed{4\text{pts}}$ ii. Find the point where this tangent line intersects the xz-plane.

$+1$ {INTERSECT XZ-PLANE $\Rightarrow y=0$

$+1$ $\left\{ 0 = 1+3u \Rightarrow u = -\tfrac{1}{3} \right.$

$\left. \begin{array}{l} x = 2+2(-\tfrac{1}{3}) = \tfrac{4}{3} \\ y = 1+3(-\tfrac{1}{3}) = 0 \\ z = 2(-\tfrac{1}{3}) = -\tfrac{2}{3} \end{array} \right\}$ $+2$

Intersection point $(x,y,z) =$ $\left(\tfrac{4}{3}, 0, -\tfrac{2}{3}\right)$

$\boxed{4\text{pts}}$ (b) Find the curvature of the curve $\mathbf{r}_2(t) = \langle t,\ t^2,\ 2t\rangle$ at $t=1$. Hint: Remember to plug in $t=1$ as early as possible. You can leave your final answer in unsimplified exact form or as a decimal rounded to 3 digits.

$+1$ $\left\{ \begin{array}{l} \vec{r}\,'(t) = \langle 1, 2t, 2\rangle \Rightarrow \vec{r}\,'(1) = \langle 1,2,2\rangle \\ \vec{r}\,''(t) = \langle 0,2,0\rangle \Rightarrow \vec{r}\,''(1) = \langle 0,2,0\rangle \end{array} \right.$

$+1$ $\left\{ \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 2 \\ 0 & 2 & 0 \end{vmatrix} = (0-4)\vec{i} - (0-0)\vec{j} + (2-0)\vec{k} = \langle -4,0,2\rangle \right.$

$+1$ $\left\{ \dfrac{|\vec{r}\,'(1)\times\vec{r}\,''(1)|}{|\vec{r}\,'(1)|^3} = \dfrac{|\langle -4,0,2\rangle|}{|\langle 1,2,2\rangle|^3} = \underbrace{\dfrac{\sqrt{16+0+4}}{(\sqrt{1+4+4})^3} = \dfrac{\sqrt{20}}{3^3}}_{+1} \right.$

$\kappa(1) =$ $\dfrac{\sqrt{20}}{27} = \dfrac{2\sqrt{5}}{27} \approx 0.166$